Ex 9 Given $\mu I \preceq \nabla^2 f \preceq L I$, $0 < \mu \leq L$ also

$T_\gamma(x) := x - \gamma \nabla f(x)$

1) $\|T_\gamma(x) - T_\gamma(y)\| \leq \max\{|1-\gamma\mu|; |1-\gamma L|\}\|x-y\|$

$$T_\gamma(x) - T_\gamma(y) = \int_0^1 \nabla T(y + t(x-y))(x-y)\,dt \Rightarrow$$

$$\Rightarrow \|T_\gamma(x) - T_\gamma(y)\| = \left\| \int_0^1 \nabla T(y+t(x-y))(x-y)\,dt \right\| \leq$$

$$\leq \int_0^1 \|\nabla T(y + t(x-y))\| \|x-y\|\,dt \leq \sup_{z \in \mathbb{R}} \|\nabla T(z)\| \cdot \|x-y\|$$

$$= \sup_{z \in \mathbb{R}} \|I - \gamma \nabla^2 f(z)\| \cdot \|x-y\|$$

also $\mu I \preceq \nabla^2 f \preceq L I$ also $\Leftrightarrow$ $\|I - \gamma \nabla^2 f(x)\| \leq |1-\gamma L|$ and $\|I - \gamma \nabla^2 f(x)\| \geq |1-\gamma\mu|$

$$\Rightarrow \|T_\gamma(x) - T_\gamma(y)\| \leq \|x-y\| \cdot \max\{|1-\gamma L|; |1-\gamma\mu|\};$$

2) if $\gamma = \frac{2}{\mu+L} \Rightarrow |1-\gamma L| = |1-\gamma\mu| = \frac{L-\mu}{L+\mu}$

also if $\gamma > \frac{2}{\mu+L} \Rightarrow \mu\gamma < \frac{2\mu}{\mu+L} \Rightarrow 1-\mu\gamma > 1 - \frac{2\mu}{\mu+L} = \frac{L-\mu}{L+\mu} \Rightarrow$

$\Rightarrow |1-\mu\gamma| > \left|\frac{L-\mu}{L+\mu}\right| \Rightarrow \max(|1-\gamma L|; |1-\mu\gamma|) > \frac{L-\mu}{L+\mu}$

$\Rightarrow$ ...

also the same if $\gamma < \frac{2}{\mu+L} \Leftrightarrow 1 - L\gamma > 1 - \frac{2L}{\mu+L} = \frac{\mu-L}{\mu+L} \Leftrightarrow$

$\Rightarrow |1-L\gamma| > \frac{\mu-L}{\mu+L} \Leftrightarrow$

$\Leftrightarrow \max(|1-\gamma L|; |1-\mu\gamma|) > \frac{L-\mu}{L+\mu}$ $\Leftrightarrow \gamma = \frac{2}{\mu+L}$ is optimal

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$$\|x^{k+1} - x^*\| = \|T_\gamma(x^k) - T_\gamma(x^*)\| \leq \frac{L-\mu}{L+\mu}\|x^k - x^*\| \quad (F)$$

$$\|x^k - x^*\| \leq \|x^0 - x^*\| \cdot \left(\frac{L-\mu}{L+\mu}\right)^k$$

In Theorem 1.5 $\|x^k - x^*\| \leq \|x^0 - x^*\| \cdot (1-\gamma\mu)^k$ where $\gamma = \frac{1}{L}$

$$\Rightarrow \|x^k - x^*\| \leq \left(1 - \frac{\mu}{L}\right)^k \|x^0 - x^*\| = \left(\frac{L-\mu}{L}\right)^k \|x^0 - x^*\|$$

and $\frac{L-\mu}{L+\mu} < \frac{L-\mu}{L}$ also $\frac{L-\mu}{L} : \frac{L-\mu}{L+\mu} = \frac{L-\mu}{L} \cdot \frac{L+\mu}{L-\mu} = \left(1 + \frac{\mu}{L}\right)$ time better

Ex 10/

If f is L-smooth and μ-strong convex then

$$\frac{1}{L}\|\nabla f(x^u)-\nabla f(x^*)\| \leq 2D_f(x^u;x^*) \leq \frac{1}{\mu}\|\nabla f(x^u)-\nabla f(x^*)\|^2$$

So (i) and (ii) are true $\frac{1}{L} \leq \gamma_u \leq \frac{1}{\mu}$

For (iii) From Lemma (14) we have

$$\|x^{u+1}-x^*\|^2 \leq (1-\gamma_u\mu)\|x^u-x^*\|^2 \Big(-\underbrace{2\gamma_u}_{\geq 0}\underbrace{(1-\gamma_u L)}_{\geq 0}\underbrace{D_f(x^u;x^*)}_{\geq 0}\Big) \quad \underbrace{\leq 0}$$

$$\Rightarrow \|x^{u+1}-x^*\|^2 \leq (1-\gamma_u\mu)\|x^u-x^*\|^2$$